SafePay

The money gets through. The abuse doesn't.

Real-time detection of abuse hidden in payment messages, with a signed record the victim can take to court. Built and measured on an IBM z17.

Payments to Sam · 01:10–01:14			9 credits
01:10:00	+$0.01	I	WITHHELD
01:10:30	+$0.01	am	WITHHELD
01:11:00	+$0.01	going	WITHHELD
01:11:30	+$0.01	to	WITHHELD
01:12:00	+$0.01	find	WITHHELD
01:12:30	+$0.01	you	WITHHELD
01:13:00	+$0.01	and	WITHHELD
01:13:30	+$0.01	kill	WITHHELD
01:14:00	+$0.01	you	WITHHELD

Read together: **“I am going to find you and kill you.”** Threat 1.00 · case P1 · money delivered · screened in milliseconds

THE PROBLEM

You can block a number. You can't block a bank transfer.

When someone escapes an abuser, they block them on every app. But anyone with their account number can send one cent with a 140-character message attached, and it lands in their banking app, often at night.

≈1,500

high-risk cases a year at Commonwealth Bank of Australia, which built a model to find them and gave it to other banks for free.

24,000

payments blocked by Westpac over abusive wording, sent by 19,000 customers.

1 → 9

one threat, nine transfers. Santander described a death threat split across payments so no single message tripped a filter.

Sources: Commonwealth Bank newsroom (2023) and arXiv 2303.08016; Westpac via BankingDay; Santander via the Financial Times and Today's Family Lawyer.

WHY TODAY'S FIXES FALL SHORT

Banks block words and score monthly. Abusers adapt in minutes.

Word lists are easy to evade

`k.i.l.l` · d3ad · look-alike letters you · 🔪 · or one word per payment. Santander said banned words "are not known to stop senders".

Monthly scoring is slow

Commonwealth Bank's open-source model scores each relationship once a month. Re-implemented on my test world, it took a median of **19 days** to flag abuse.

Menace can be polite

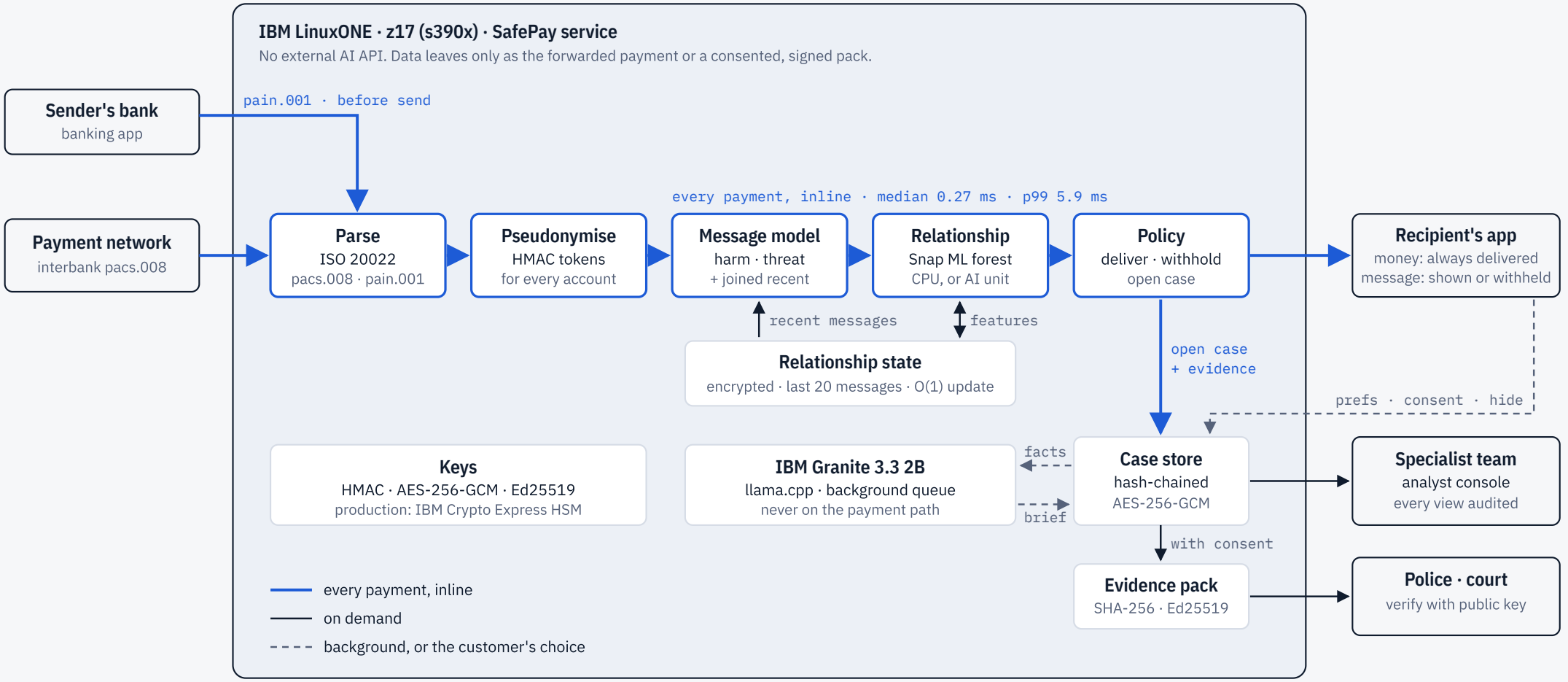
“Saw you at the gym today” scores **0.02** on a toxicity model. The danger is in the pattern: penny payments at night, never answered, after being blocked.

Blocking money hurts victims

The payment may be child support or rent she depends on. The message is the harm, not the money.

HOW SAFEPAY WORKS

One decision per payment, inline, on one IBM z17.



Blue: every payment, inline. Money always reaches the recipient; only the message can be withheld. Data leaves the z17 only as the forwarded payment or a consented, signed pack.

DETECTION

Three layers, because each one alone misses something.

1 · Message model

Harm and threat heads trained on 260,000 real comments. Undoes evasion tricks first, recovering 95% of the recall they remove. Identity terms are masked so “gay” or “muslim” isn't read as abuse.

“…worthless b1tch” → **harm 1.00**

2 · Split messages

Joins the sender's references from the last 30 minutes and scores the sentence they form. Words under four letters are judged only in context.

9 words, each ≈ 0.0 → **threat 1.00**

3 · Relationship model

23 features kept per sender→recipient pair and updated in O(1): penny payments, night share, replies, bursts, an escalation detector, contact after being hidden.

“saw you at the gym” 0.02 → **pattern 1.00**

IBM GRANITE 3.3 runs locally on the z17: a brief for analysts, and an independent second opinion that never sees the model's score. It stays advisory (slide 10).

THE VICTIM'S SIDE

Protect her now. Help her prove it later.

Protect

- A neutral **“Message settings”** tab: how the bank may contact her, at what times, and whether someone else sees her phone. Encrypted.
- **Quick exit:** one tap swaps the screen for an ordinary account view.
- **Protect after hide:** for 14 days, brand-new senders are hidden by default, so an abuser who switches account is withheld from his first message.
- Support lines (1800RESPECT, 000) and a confidential in-app line.
- Analysts see her safe-contact rules first, before anything else.

Prove

- With her consent, per sender and purpose (police, protection order, legal advice), and withdrawable.
- A **signed evidence pack**: every payment from the first message, including those before the case opened. Facts come from the log, not AI.
- SHA-256 hash-chained evidence log, and an Ed25519 signature anyone can check.

```
This record: VALID. Signed by the bank, log intact.
```

```
Copy with one full stop added: NOT VALID.
```

RESULTS · REAL-TIME VS MONTHLY BATCH

Nearly as many caught, and over two weeks sooner.

3.3 days

median time from first abusive payment to a case

monthly batch: 19.3 days

92.0 %

of abusive messages withheld as they arrived

monthly batch: 0% (it acts after the month)

452 / 481

abusive relationships caught

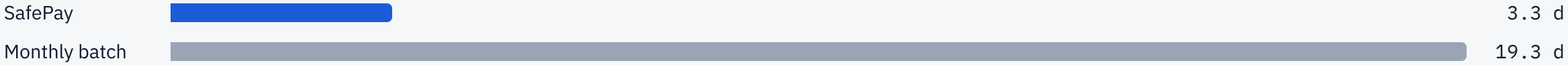
monthly batch: 465 / 481

0

false alarms in 6,390 innocent relationships

monthly batch: 2

Median days from the first abusive payment to a case (bars to scale)



Held-out test world: 6,871 relationships, 44,426 payments, real Civil Comments test-split text, never used for any decision. Weak spot: abusive child-support payers, paid fortnightly or monthly, take a median of 58 days to reach a case. Zero false alarms here reflects a synthetic world; the next slide tests it at scale.

STRESS TEST · I MADE MY OWN WORLD HARDER

Harder data exposed two weaknesses. One fix was the model, one was policy.

Realistic people

People now share relationships, and innocent people hide senders too. The previous model had learned **“a hide means abuse”** and flagged friends who fell out over “groceries split”: 9–12 false alarms per 4,800 innocent.

`Retrained → 0 false alarms`

Account switching

Hidden by the victim, the abuser returns from a new account or a friend's. Caught 84% in 40 hours, but 28% of his messages got through first. Teaching the model about the recipient's side didn't help.

Protect after hide: 14 days of default hiding for new senders → **99% withheld**, 0.3% of innocent relationships see a soft hide.

A realistic share of abusers

In tests 7% of relationships are abusive. Real banking is far rarer, so I replayed 305,163 innocent relationships, 1.7 million payments.

`15 false alarms · 0.49 per 10,000`

If 1 in 1,000 relationships were abusive, 95% of cases would be real; at 1 in 10,000, 66%. A false alarm costs a human review and a soft hide, never a blocked payment.

OUTSIDE CHECK · COMMONWEALTH BANK'S PRODUCTION MODEL

A model trained on real abuse recognises ours, and misses the quiet kinds.

CBA's open-source model was trained on real, expert-labelled cases and never saw my data. On my simulated world it scores **ROC AUC 0.87–0.94 per month**, against 0.80 on real data.

- My synthetic abuse looks like real abuse to it.
- My world is easier than reality, so I don't claim real-world precision.
- SafePay was trained on this simulator and CBA's model wasn't. Part of SafePay's lead is home advantage. (Run on the previous simulator and model.)



RESPONSIBLE AI

I tested my own work, and kept the failures on record.

Fairness

My first model hid harmless messages just for naming a group. On Jigsaw's identity templates:

Harmless sentence mentions	Before	After masking
“gay”	99.7%	13.2%
“black”	88.6%	13.2%
“muslim”	76.2%	13.2%

13.2% is now the rate for any template sentence. Toxic templates caught rose from 80.7% to 97.4%.

Honest evaluation

- Version 1 memorised hand-written phrases (precision 70%). I split phrases three ways and report the final model on an untouched world.
- Granite's second opinion was measured, not assumed: +5 caught for +1 false alarm on a fresh world, but +1 for +2 on another. So it advises and never decides.
- I sped Granite up 5× to 11 s per opinion, by reading one token's probability instead of generating a sentence.
- A harder world exposed a shortcut: the model had learned “a hide means abuse” and flagged friends who fell out: 9–12 false alarms per 4,800 innocent. Retrained: 0.

WHY IBM Z · MEASURED ON AN IBM Z17, MACHINE TYPE 9175

Screen payments where they already run, without data leaving.

933 /s
payments screened in one process, p99 5.9 ms

577 /s
over HTTP with real ISO 20022 XML, p99 6.7 ms

11 µs
relationship model on Snap ML, CPU engine

0
external AI APIs. Granite runs locally, big-endian.

- **On-chip AI accelerator, benchmarked honestly.** For this small forest, one payment at a time, Snap ML on the Telum AI unit took 61–97 µs against 11 µs on CPU, so I ship the CPU engine. In the lab container, the accelerator ran a dense network 11× faster than CPU at batch 2048.
- **Data stays on the platform.** AES-256-GCM on all stored message text, HMAC account tokens, and Ed25519-signed packs. In production the keys belong in Crypto Express HSMs.
- **Native s390x stack.** Ubuntu's numpy, SciPy and scikit-learn, IBM Snap ML and zDNN, llama.cpp for z16. Packaged as a Podman container.
- **Portability findings.** H2O's model scorer misreads trees on big-endian (feature 33 read as 8448), and GCC 13 can't target the z17. Both are documented for banks moving models onto Z.

SCALABILITY AND FEASIBILITY

Built on the payment standard banks already use.

- **Plugs into existing flows.** ISO 20022 in, decision and forwarded pacs.008 out, at either bank.
- **Scales out.** Each relationship's state is independent and updated in O(1), so traffic shards by sender across processes or LPARs.
- **Learns.** Analyst confirm/dismiss labels are already stored for retraining.
- **Next:** pilot on a bank's consented, pseudonymised logs; other languages via Granite; keys in HSMs; other rails such as UPI.

Real	Simulated, and why
ISO 20022 parsing and the rewritten message	The bank: no hackathon can connect to one
1.14 million real comments (Civil Comments, CC0)	Abusive relationship timelines: public data of real victims shouldn't exist
Models, Snap ML and Granite on the z17	Names and accounts
Whatever you type in the live demo	

SafePay

The money gets through, the abuse doesn't, and the evidence does.

Code	`github.com/suhasadiga06/IBMZ`
Live demo	`http://148.100.76.164:8090` User `judge`, password in the submission. Sender, screening trace and recipient's app, side by side
Analyst console	`/analyst` · Ranked cases, safe-contact rules, Granite brief, evidence packs
Evidence page	`/system` · Every number in this deck, read live from the result files
Built by	Suhas Adiga, individual entry